Question	Answer	Marks	Guidance
1(a)	Data has outliers / is skewed / lacks symmetry, suggesting population/distribution may not be symmetrical.	**B1**	Must be in context. Condone no explicit mention of population/distribution or a definite statement relating to population.
		1	
1(b)	H_0: population median is [$]32,500. H_1: population median is not [$]32,500.	**B1**	Allow m, but not μ. If m defined, must see **population.** Median not mean or average.
	Test statistic = 4 (or 11)	**B1**	Can be implied by $P(X \leqslant 4)$ or $P(X < 4)$. (For use of Normal approximation, the value of the test statistic can be implied by 4.5 or 3.5 for 4 and 10.5 or 11.5 for 11).
	[$X \sim B(15, 0.5)$] $P(X \leqslant 4) = 0.0592$ OR $P(X \geqslant 11) = 0.0592$	**B1**	AWRT 0.0592. Might see 0.1184/0.1185/0.118/0.119 from $2P(X \leqslant 4)$ or $2P(X \geqslant 11)$. Use of N(7.5,3.75) scores B1 for ±1.549.
	$'0.0592' > 0.05$, accept H_0 / do not reject H_0 / not significant.	**M1**	Compare *their* 0.0592 (must be a tail probability) with 0.05 (or 0.1 if 1-tail test used) OE and appropriate result (may be in terms of H_1). Result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses. Use of Normal: compare *their* ±1.549 with ±1.645, signs must be consistent (1.282 for 1-tail test).
	Insufficient evidence to reject [company's] claim. OR Insufficient evidence to suggest that median [salary] is not [$]32,500.	**A1**	Correct conclusion from correct working ignoring hypotheses. In context. Level of uncertainty in language is used (for example, not 'prove'). Allow "Not enough evidence / no sufficient evidence that / to show / conclude that…". Do not accept "No evidence…". Do not accept statements such as "there is sufficient evidence to suggest…".
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